Programming Assignment 1 | Team 14

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Due Wednesday, February 25, 12:50 PM on Canvas

Software Design

The software design follows a client/server paradigm. The server is run first, on a port specified via the command line. The client is then run on a separate machine. The client must first sepcify the server's Wireless LAN's IPv4 address (accessible on server via command 'ipconfig') and then the port the server specified. A TCP/IP connection over the specified port is established for the communication to occur. Communication between the two was programmed according to protocol outlined in assignment.

Timing Statistics

Statistic	Time (ms)
Mean	26.4
Median	17
Mode	17
Minimum	1
Maximum	74
Standard Deviation	23.75

Lessons Learned

Many important lessons were learned during the research and programming for this assignment. One such lesson was the need to flush ObjectOutputStreams. Characters can hide out in a buffer, evading being sent to the other machine. This can cause bugs that are hard to diagnose. Another lesson was about how addressing for computers work. To communicate with another device, its IP address needs to be known in addition to the port expecting the message.